

Causal Discovery from Complex Survey Data

A Survey-Weighted Ordinal Bayesian Network Approach

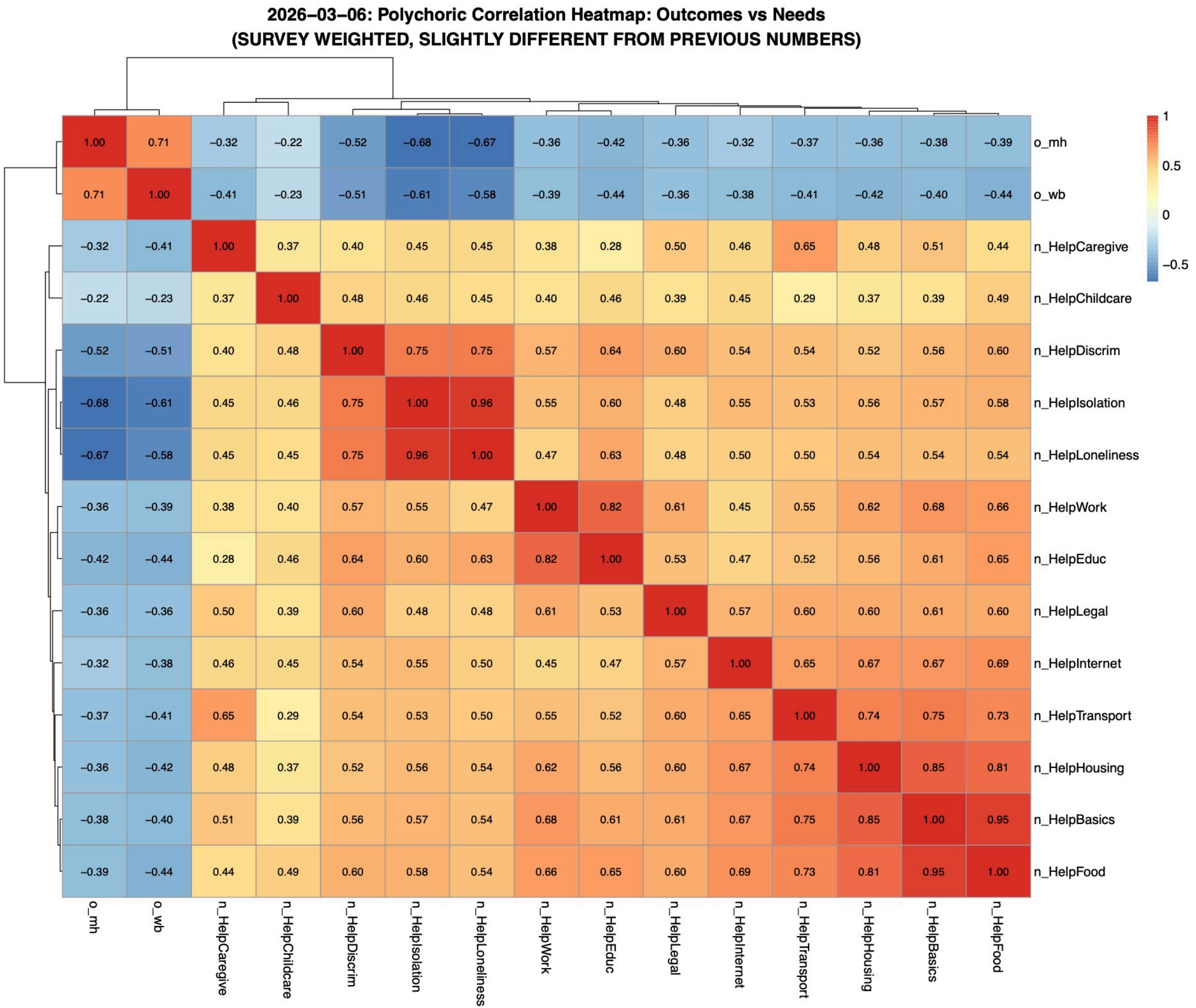
Why Causal Discovery?

- Correlation heatmap: **everything correlates with everything**
 - Need items intercorrelate 0.29–0.96; all negatively with outcomes (-0.22 to -0.68)
 - Standard regression → “all needs are associated with poor outcomes” → **clinically useless**
- Policy requires **directional** knowledge:
 - If VA addresses basic financial needs, does discrimination/isolation burden decrease downstream?
 - Does intervening on work/education improve well-being directly, or only through an intermediate pathway?
- Existing VA/SDOH literature is **entirely associational**
 - Regression models assume causal structure; they don’t discover it

Note

- Building a network to see which needs are more upstream vs. more downstream with the health outcomes (i.e., well-being and mental health)
- A network adjusted for age, sex, and race/ethnicity
- Target deliverable: tree/network structure

Why Causal Discovery? (notes on next page)



From 15 Items to 10 Nodes

- 15 raw CAHPS need items $\rightarrow \sim 4.2 \times 10^{18}$ possible DAGs
 - Computationally intractable for algorithm
 - Individual items too granular for interpretable causal claims
- **Polychoric correlations justify clustering: $\rho \geq 0.75$**
 - HelpBasics / HelpFood / HelpHousing: $\rho = 0.85\text{--}0.95$
 - HelpWork / HelpEduc: $\rho = 0.82$
 - HelpDiscrim / HelpIsolation / HelpLoneliness: $\rho = 0.75\text{--}0.96$
- 5 standalone items don't cluster: Caregiving, Childcare, Transport, Internet, Legal

Note

- Factor in that these are ordinal variables and coming from survey data (so we need survey weighting): 3 levels of need
- We previously considered 0.7 as a cut point; however, for today's discussion, we used 0.75 as the cut point
- Clusters: (1) basics/food/housing; (2) work/education; and (3) discrimination/isolation/loneliness
- 5 standalone items: (1) caregiving; (2) childcare; (3) transportation; (4) internet; and (5) legal

Final Node Set (clustering if $\rho \geq 0.75$)

Node	Type	Levels	Source
Basic Needs	Composite	3	PayBasics + LackFood + HaveHousing
Work/Education	Composite	3	FindWork + LackEduc
Discrim./Isolation	Composite	3	Discrim + Isolation + Loneliness
Caregiving	Item	3	HelpCaregive
Childcare	Item	3	HelpChildcare
Transport	Item	3	HelpTransport
Internet	Item	3	HelpInternet
Legal	Item	3	HelpLegal
Well-being	Outcome	4	o_wb
Mental Health	Outcome	5	o_mh

Assumption Diagnostics

- **Proportional odds (Brant test)**
 - Minor violations, nothing systematic
 - Consistent with large-sample power issue, not genuine PO failure
 - Post-analysis Brant on discovered parents confirms
- **Survey weights**
 - Report: n , n_{eff} , design effect, weight range
 - No single observation dominates the pseudo-likelihood
- **Sparse cells**
 - No degenerate cross-tabulations
 - All pairwise combinations have adequate cell counts

Note

- Using a multinomial model to look at directionality
- Assumptions we've made so far seem okay (Soumik will do some more sensitivity analysis mostly in anticipation of reviewer comment/feedback)

The swa-oBN Framework

- **Survey-weighted pseudo-BIC** corrects for sampling design
 - Pseudo-log-likelihood targets population, not sample
 - Kish effective sample size in penalty term
- **Ordinal regression** breaks Markov equivalence
 - Cumulative link model → edge direction identifiable for $L \geq 3$
 - Multinomial BNs cannot orient edges; the oBN can
- **Two-step scoring** (new contribution)
 - Add/delete → penalized pseudo-BIC (skeleton discovery)
 - Reversal → unpenalized pseudo-likelihood (edge orientation)
 - Eliminates finite-sample bias from heterogeneous ordinal levels
- **Covariate adjustment** (Z matrix): age, race/sex
 - Included in every node regression, excluded from DAG

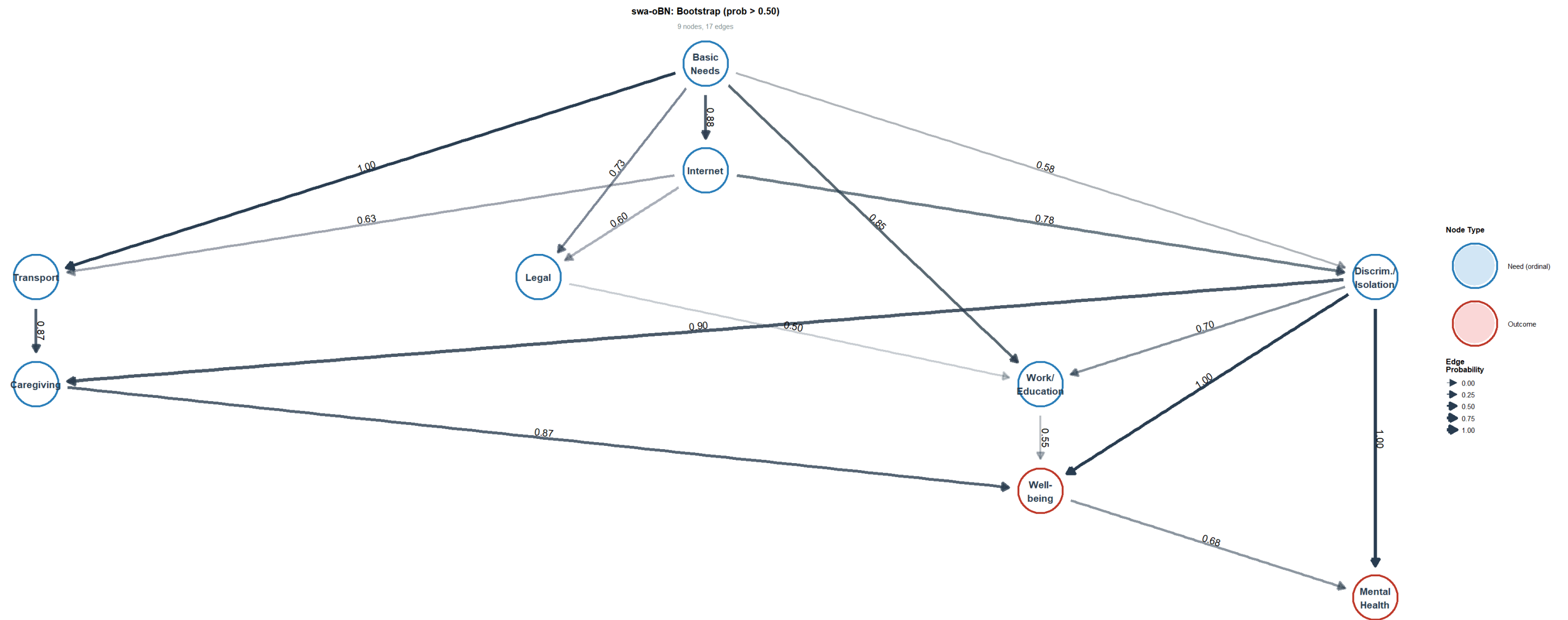
Expert Constraints

- **Blacklist:** outcomes cannot cause needs
 - Well-being →/any need variable
 - Mental Health →/any need variable
 - Encodes minimal temporal/substantive assumption
- **No whitelist** — all need→need and need→outcome edges discovered from data
- **Confounder adjustment** via Z matrix (not DAG nodes): Age (categorical) and Race × Sex (categorical)

Note

- Constraint: We won't allow for an arrow from an outcome to a need

Estimated Causal Network



i Note

- Covariate adjusted network.
- Updated network includes “edge probability”
- Bootstrap group of networks
- The thicker/bolder the line, the more confident we are in the arrow between the two nodes (score will be between 0.0 and 1.0)
- We may not want to consider an edge <0.5

Question: what threshold is a good cutoff for this type of network?

Finding 1: Basic Needs Is the Upstream Hub

- Direct edges to **6 of 8 other nodes**
 - Transport (1.00), Work/Education (0.85), Internet (0.88)
 - Discrim/Isolation (0.58), Caregiving (0.63), Legal (0.73)
- Unmet basic financial needs are a **direct cause of six downstream variables**
- **Policy implication:** addressing basic needs may cascade benefits downstream

Finding 2: Discrim/Isolation → Outcomes

- Most stable edges in the entire graph:
 - Discrim/Isolation → Well-being: **1.00**
 - Discrim/Isolation → Mental Health: **1.00**
 - Discrim/Isolation → Work/Education: **0.70**
- Social isolation and perceived discrimination are the **proximal causal mechanism** linking needs to outcomes
- Not merely correlated — the DAG identifies them as the **direct pathway**
- **Policy implication:** interventions that don't address isolation/discrimination may miss the primary channel to health outcomes.

Finding 3: Mental Health Is Mediated (not very sure of message here...will think more.)

- Mental Health does **not** receive many direct edges from need variables
- Primary pathways:
 - Needs → Discrim/Isolation → Mental Health (1.00)
 - Needs → Discrim/Isolation → Well-being → Mental Health (0.68)
 - Work/Education → Well-being (0.55) → Mental Health
- **Two-step structure:** needs don't directly damage mental health — they do so through isolation and degraded well-being
- **Policy implication:** mental health interventions targeting needs alone, without addressing the isolation pathway, may be insufficient.

Limitations

- **Causal sufficiency untestable** — unmeasured confounders possible
 - Disability rating, combat exposure, neighborhood effects
 - Z matrix (age, race/sex) mitigates but cannot eliminate
- **Binary items:** edge direction from BIC penalty only, not ordinal machinery
 - Composite↔composite and composite→outcome edges are most trustworthy
- **Naive bootstrap**, not design-appropriate (Rao-Wu)
 - Intermediate inclusion probs (0.50–0.85) interpreted with caution
- **Cross-sectional data** — DAG is contemporaneous, not temporal
 - Blacklist encodes minimal temporal assumption

Summary

1. **Basic Needs** is the root cause hub — addressing it cascades downstream
2. **Discrimination/Isolation** is the primary pathway to health outcomes
3. **Mental Health** is reached through mediation, not directly from most needs
4. **Work/Education** is downstream — not an independent root cause
5. **Survey weighting matters** — ignoring it produces spurious causal structure
6. **swa-oBN** provides the first formal guarantees for causal discovery from complex survey data

Note

Findings: In addition to Soumik's takeaways, we may want to include information in discussion about connection between caregiving and well-being yet caregiving not a screening priority in routine practice (e.g., not a measure in ACORN, Accountable Health Communities (AHC))